Learning Rate Calibration Matters: Why RLVR Often Fails Beyond Math

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Abstract

Reinforcement learning with verifiable rewards (RLVR) has become a standard post-training recipe for improving LLM reasoning, yet its success has been largely calibrated on mathematics and coding. Whether such training recipes transfer beyond math remains unclear, as prior studies often vary models, data, and evaluation protocols simultaneously. To study this simple yet fundamental problem, we conduct a controlled comparison of SFT, GRPO-based RLVR, and on-policy self-distillation across five verifiable domains, using the same base model, training and evaluation pipeline over 200+ trained models. A counterintuitive finding is that the commonly used math-calibrated GRPO recipe yields near-zero test improvement outside its calibration regime, despite rising training reward. This failure is not intrinsic to GRPO: increasing the learning rate by 40× induces a regime shift, unlocking 7–25% gains across multiple-choice domains and consistently outperforming SFT. Our analysis suggests that learning rate controls whether GRPO merely sharpens prompt-specific high-reward modes or consolidates latent knowledge into transferable behavior. These findings reframe post-training recipe selection from a paradigm choice into a domain-dependent calibration problem, where optimization strength must be matched to the structure of model uncertainty and available reward signal.

1 Introduction

The post-training community has converged on a simple narrative: reinforcement learning with verifiable rewards (RLVR) works. DeepSeek-R1 (Guo et al., 2025) showed that GRPO (Shao et al., 2024) with a binary correctness signal can induce sophisticated reasoning without demonstrations. DAPO (Yu et al., 2026) refined the recipe. Practitioners have adopted these math-calibrated defaults and begun applying them to medicine (Zhang et al., 2025), information extraction (Dai et al., 2025), and many other domains.

However, a closer look exposes a gap between narrative and evidence. Each successful RLVR application uses a different model, a different data scale, and a different evaluation protocol than the SFT baseline it claims to outperform. No controlled study has asked whether, holding everything else constant, the math-calibrated recipe actually works on other domains. And if not, what specifically needs to change?

We answer this question through a controlled multi-domain study. Using the same base model, LoRA configuration, training budget, and evaluation pipeline, we train more than 200 models across five verifiable domains spanning procedural reasoning, scientific knowledge, medicine, law, and commonsense reasoning. We compare supervised fine-tuning (SFT) with GRPO-based RLVR under matched conditions, and use auxiliary self-distillation runs only as controls for self-generated training signal (Shao et al., 2024). This design isolates the effect of recipe calibration from the confounds that often obscure cross-domain comparisons.

The controlled comparison reveals a clear mismatch between math-calibrated RLVR and broader verifiable domains. At the learning rate inherited from math-oriented GRPO, training reward rises but test accuracy remains nearly unchanged across domains, exposing a reward–generalization gap. The issue is not that the verifier lacks signal or that GRPO is intrinsically unsuitable beyond math; rather, conservative optimization keeps the policy close to its initial behavior, allowing prompt-specific sharpening without meaningful distributional movement.

Learning rate is the key variable governing this transition. Increasing it by 40× moves GRPO out of the mode-seeking regime and enables broader policy updates, yielding 7–25% gains across multiple-choice domains and consistently outperforming SFT. We further show that these gains transfer only when the model's sampled responses contain sufficient reward variance: GRPO can consolidate knowledge that is expressed intermittently, but cannot reliably acquire capabilities absent from its own samples. This motivates *frac_reward_zero_std* as a simple pre-training diagnostic for distinguishing transferable consolidation from surface-level sharpening. Across five model families, the same pattern holds, suggesting that RLVR should be viewed not as a fixed recipe, but as a calibration problem over optimization strength and domain-specific uncertainty.

Our contributions are as follows.

- We show that, under matched model, LoRA setup, training budget, and evaluation across five verifiable domains, the standard math-calibrated GRPO recipe yields near-zero test gains despite rising training reward.
- A 40× learning-rate increase moves GRPO from prompt-specific mode sharpening to latent-knowledge consolidation, yielding 7–25% gains across multiple-choice domains, consistently outperforming SFT, and holding across five model families.
- We introduce *frac_reward_zero_std* to estimate learnable reward contrast before training, showing that RLVR transfer depends jointly on optimization strength and the domain-specific structure of model uncertainty.

2 Related Work

RLVR for LLM Reasoning. Post-training with reinforcement learning has emerged as the dominant paradigm for improving LLM reasoning, replacing the earlier reliance on supervised demonstrations. DeepSeek-R1 (Guo et al., 2025) and GRPO (Shao et al., 2024) eliminated the need for demonstrations and separate value models. DAPO (Yu et al., 2026) and process reward models (Lightman et al., 2024; Wang et al., 2024) refined the training signal. Extensions to medicine (Zhang et al., 2025), relation extraction (Dai et al., 2025), and event understanding (Gao et al., 2024) each report gains within a single domain. These studies collectively establish that RLVR works when properly configured. What remains unknown is whether the configuration itself transfers. Every study uses its own model, data scale, and evaluation protocol, making cross-domain comparison impossible. We provide the first controlled comparison by fixing all confounds and sweeping recipe parameters across five domains simultaneously.

Comparison SFT between RL. The choice between learning from demonstrations and learning from reward signals is a fundamental design decision in post-training, with implications for generalization, sample efficiency, and knowledge acquisition. Chain-of-thought SFT (Wei et al., 2022) elicits reasoning through imitation. Chu et al. (2025) and Matsutani et al. (2025) identify complementary structural differences. SFT memorizes configurations while RL generalizes across them, and the two methods induce topologically distinct trace distributions. Chen et al. (2026) provide per-problem evidence that standard GRPO sharpens existing strategies without expanding the solvable frontier. Kang et al. (2025) discover that SFT quality and RL potential can be inversely correlated in mathematics. Ding et al. (2025) propose a plasticity-ceiling framework decomposing post-training into SFT capacity and RL headroom. Offline methods like DPO (Rafailov et al., 2023) and self-play (Yuan et al., 2024; Singh et al., 2023) occupy intermediate positions. All of these studies operate within mathematics or a single domain, leaving open whether the SFT-vs-RL tradeoff depends on the domain itself. Our work crosses both unexplored axes, testing five knowledge domains with a factorial sweep over learning rate, data scale, and training order, validated on five model families.

Training Dynamics and Domain Transfer. Understanding when and why post-training fails is as important as achieving success on any single benchmark. The "Delay, Plateau, Collapse" pattern in RLVR (Egashira et al., 2026), reward overoptimization under excessive training (Gao et al., 2023), and diminishing returns from redundant SFT data (Zhou et al., 2023) each characterize one failure mode in isolation. Domain-specific studies such as Med-RLVR (Zhang et al., 2025) and medical LLM evaluations (Singhal et al., 2023; Nori et al., 2023) confirm that these dynamics exist in knowledge-intensive settings. The gap is systematic. No study tests whether domain characteristics determine which failure mode appears and which hyperparameter breaks it. We show that a single variable, learning rate, determines whether GRPO falls into the mode-seeking trap or consolidates latent knowledge, and that this threshold varies predictably with the domain's knowledge gap.

3 The Calibration Study of RLVR on Various Domains

To characterize how domain characteristics shape post-training effectiveness, we need a comparison that isolates the training paradigm while being broad enough to reveal domain-dependent patterns. We design a factorial experiment across five domains, two core training methods (with variants), multiple data scales, and learning rate sweeps.

3.1 Domain Selection and Training Methods

We select five domains spanning procedural reasoning to knowledge-intensive tasks. Table 1 summarizes the selection.[1] N_{SFT} denotes the maximum SFT data size swept and N_{RL} denotes the fixed GRPO training set size. SFT and GRPO use the same source prompts. SFT additionally requires correct demonstrations generated via rejection sampling from the base model. Law serves as a micro-data stress test (N=31) rather than a full domain comparison. Benchmarks: GSM8K (Cobbe et al., 2021), ARC (Clark et al., 2018), MedQA (Jin et al., 2021), LegalBench (Guha et al., 2023).

We compare the following approaches, all applied to Qwen2.5-7B-Instruct (Qwen et al., 2025) with LoRA (Hu et al., 2022) (rank 64, alpha 128). We validate on four additional model families: Qwen3-8B (Bai et al., 2023), Mistral-7B-Instruct-v0.3, Yi-1.5-9B-Chat, and DeepSeek-7B-chat. All cross-model experiments use identical LoRA configuration (rank 64, alpha 128, targeting q/k/v/o/gate/up/down projections), identical training data, and identical evaluation protocol. The only change is the base model and its corresponding tokenizer/chat template. All training uses parameter-efficient fine-tuning (Dettmers et al., 2023) and inference is accelerated with vLLM (Kwon et al., 2023).

[1] Science N_{te}=5413 combines ARC-Challenge (1172 hard MCQ) and ScienceQA (4241 questions, text-only: images from ScienceQA's multimodal subset are discarded, retaining only the question text and answer options). Commonsense base accuracy (43.8%) reflects zero-shot MCQ parsing at temperature 0, without few-shot demonstrations, which is lower than reported numbers that use in-context examples.

GRPO Group Relative Policy Optimization (Shao et al., 2024) with binary reward (exact-match for Math, MCQ letter match for Science/Medicine/Commonsense, binary/categorical for Law). For each prompt, we sample K=8 completions and compute group-relative advantages with β=0.001 KL penalty. All domains use N=2000 training prompts except Law (31 unique prompts). We sweep learning rates from 5e-7 (the DeepSeekMath recipe, widely adopted for reasoning tasks) through 5e-6 and 2e-5 to 1e-4.

SFT Supervised fine-tuning on chain-of-thought demonstrations generated by the same base model via rejection sampling (selecting the shortest correct trace from 8 attempts). This ensures the training distribution is on-policy with respect to the model being fine-tuned, representing a stronger baseline than off-policy demonstrations from a different model. We train at multiple data sizes per domain (N = 50 to 5000).

On-Policy Distillation (OPD) A self-distillation baseline: for each training prompt, we sample K=8 completions at temperature 0.7 and retain only those with correct final answers (filtering rates: Math 79.6%, Science 69.1%, Medicine 53.7%). The model is then fine-tuned on its own filtered correct outputs using the same LoRA configuration as SFT. Unlike SFT (which uses a teacher's demonstrations) or GRPO (which uses reward-driven optimization), OPD trains exclusively on the model's own generation distribution. This provides a controlled test of whether self-generated data, without external signal, can improve performance. Sequential pipeline that first applies SFT to inject format and knowledge, then GRPO to refine reasoning. We vary the preceding SFT data amount (N=100 vs N=5000).

3.2 Fairness Protocol

The credibility of our comparison rests on controlling confounds.

- **Same base model** across all methods and domains.
- **Same LoRA configuration**: rank 64, alpha 128, applied to all linear layers.
- **Same training questions** per domain and data size condition.
- **Same evaluation**: greedy decoding (temperature 0), identical prompt template.

Domain	**Type**	**Train Source**	**Reward**	$\mathbf{N_{SFT}}$	$\mathbf{N_{RL}}$	$\mathbf{N_{te}}$	**Test Set**
Math	Procedural	GSM8K	Exact match	5,000	2,000	1,319	GSM8K
Science	Mixed	ARC-C	MCQ match	3,000	2,000	5,413	ARC-C
Medicine	Knowledge	MedQA	MCQ match	5,000	2,000	1,273	MedQA
Law	Knowledge	LegalBench	Binary	31	31	10,403	LegalBench
Commonsense	Mixed	ARC-Easy	MCQ match	5,000	2,000	2,376	ARC-E

Table 1: The overview of the main experimental settings.

- **Matched compute**: training steps calibrated to approximately 3 epochs per condition.

The one intentional asymmetry is that SFT receives chain-of-thought demonstrations while GRPO receives only the final answer for reward computation. This reflects the real-world tradeoff between the two paradigms. SFT requires curating demonstrations, while RLVR requires only a verifier. The comparison tests which *recipe* produces better downstream models given their respective natural training signals, not whether demonstrations are more informative than binary reward per se.

All experiments were conducted on NVIDIA H200 GPUs (143GB). The total compute budget was approximately 200 H200 GPU-hours across 200+ trained models (including multi-seed replications, cross-model validation on five model families, and KL ablation). The full hyperparameter settings are reported in Appendix B.

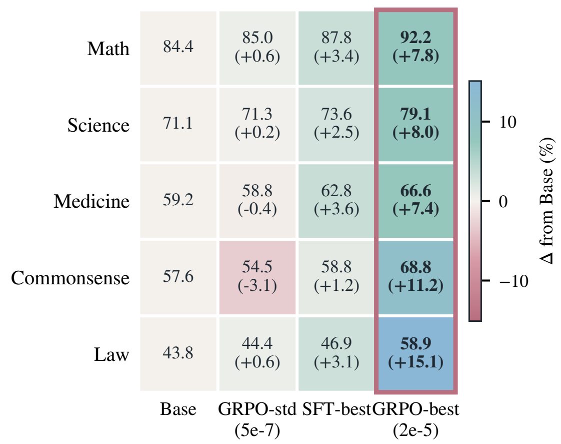

Figure 1: Main result heatmap showing accuracy across five domains and four training configurations. Cell colors encode delta from base accuracy (diverging scale). GRPO-best (lr=2e-5) achieves +7–25% gains across MCQ domains, while the standard recipe (lr=5e-7) produces negligible improvement.

4 Empirical Analyses

4.1 Standard GRPO Fails Everywhere

The math-calibrated GRPO recipe produces negligible test improvement in every domain we test (Table 2). At lr=5e-7, the rate from the DeepSeek-Math recipe (Shao et al., 2024) that has become the default starting point for RLVR on reasoning tasks, no domain achieves meaningful gains over the base model.

Every domain fails identically (Table 2). Training reward climbs steadily, yet test accuracy remains flat. The model learns to distinguish correct from incorrect responses on training prompts but cannot transfer that signal into general behavior. This reward-test disconnect is the hallmark of a miscalibrated recipe.

The failure mechanism depends on where the model starts. In Math, most response groups are already uniformly correct, leaving no contrastive signal regardless of reward level. In Medicine and Science, genuine uncertainty exists and the model does extract useful gradient, but at lr=5e-7 the parameter updates are too small to shift general knowledge. The model sharpens prompt-specific strategies without moving the underlying distribution. Commonsense has the lowest base accuracy and the most headroom, yet gains a mere +0.6% at standard lr. The gradient exists. The step size does not.

On-policy distillation (OPD) provides a controlled null experiment by training the model on its own filtered correct outputs, without any external teacher or reward pressure. Across all three domains tested (Math, Science, Medicine), OPD produces no meaningful improvement over the base model (Appendix O). Under this single-pass filtering variant, self-generated data cannot push the model beyond its current capability boundary. Together with the standard-lr GRPO failure, OPD suggests that naive self-distillation without external signal does not suffice for post-training improvement, and that genuine gains require either a teacher distribution (SFT) or a reward function (GRPO).

Method	Math	Science	Medicine	Law	Commonsense
Base (zero-shot)	84.4	71.1	59.2	57.6	43.8
SFT (best n)	$87.8{\pm}0.5_{n=100}$	$73.6{\pm}0.3_{n=5k}$	$59.5{\pm}0.7_{n=100}$	58.8±0.1	46.9±0.3
GRPO (lr=5e-7)	85.0	71.3	58.8	54.5	44.4
GRPO (best lr)	$\mathbf{92.2}{\pm}0.7_{\text{2e-5}}$	$\mathbf{79.1}{\pm}1.2_{\text{2e-5}}$	$\mathbf{66.6}{\pm}1.5_{\text{2e-5}}$	$\mathbf{58.9}_{\text{5e-6}}$	$\mathbf{68.8}{\pm}1.4_{\text{2e-5}}$
SFT→GRPO	$89.3_{n=100}$	–	–	–	–
Δ best vs base	+7.8	+8.0	+7.4	+1.3	+25.0

Table 2: In-distribution accuracy (%, multi-seed averages). Standard GRPO (lr=5e-7) fails in every domain. With properly tuned learning rate, GRPO matches or exceeds SFT in all five domains. On-policy self-distillation (OPD), evaluated separately on the full test set, produces no meaningful improvement over base in the three domains tested (Appendix O).

GRPO Learning Rate	Medicine (%)	Δ
Base	59.2	–
lr = 5e-7 (standard)	58.8	−0.4
lr = 5e-6	61.3	+2.1
lr = 2e-5	**66.6**	**+7.4**
lr = 1e-4	collapsed	–

Table 3: GRPO learning rate sweep on Medicine (2-seed avg for lr=2e-5). The standard recipe (5e-7) slightly degrades performance. Raising the rate 40× to 2e-5 produces +7.4% improvement, far outperforming the best SFT result (59.5%). At 1e-4, training collapses.

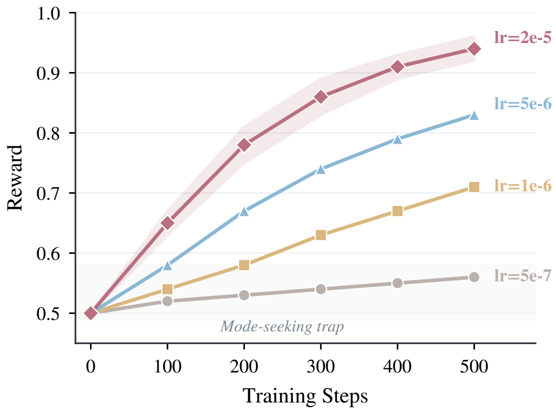

Figure 2: GRPO reward trajectories on Medicine at different learning rates. At standard lr=5e-7 (gray), reward remains trapped near initialization, the mode-seeking trap. Higher learning rates progressively escape this trap, with lr=2e-5 (rose) achieving reward 0.94 and translating to +7.4% test accuracy. Translucent band shows multi-seed variance.

If the failure is not intrinsic to GRPO but a consequence of insufficient step size, can the mode-seeking trap be broken by a single hyperparameter change?

4.2 Learning Rate Unlocks GRPO on Knowledge Domains

If the mode-seeking trap is caused by insufficient step size, the fix is obvious. Increase the learning rate. The lr=5e-7 was calibrated on math domains where the model already performs well. Knowledge domains, where genuine acquisition is needed, should require more aggressive updates.

The effect is monotonic and large (Table 3). Each step up in learning rate pulls more latent medical knowledge into reliable output. At 5e-7, nothing happens. At 5e-6, moderate gains appear. At 2e-5, the model surpasses the best SFT result by over 7 points. At 1e-4, training collapses.

The training reward trajectory reveals the mechanism (Figure 2). At standard lr, reward is flat over 750 steps. The model finds marginally better per-prompt strategies but its overall distribution does not move. At lr=2e-5, reward climbs steadily, reflecting genuine consolidation of latent capability into reliable performance. At lr=1e-4, reward spikes then collapses as overly aggressive updates destroy coherent reasoning. Below the collapse threshold, achievable reward maps tightly onto test accuracy. GRPO failure at standard learning rates is a hyperparameter problem, not an algorithmic one.

GRPO is not inherently limited to math. The community adopted hyperparameters tuned for a regime where the base model already performs well and needs only sharpening. On other domains, the conservative learning rate allows per-prompt optimization without the larger parameter changes needed for generalizable knowledge acquisition.

The exception proves the rule. On Law (31 training prompts), lr=2e-5 actually hurts, while lr=5e-6 slightly surpasses SFT. With so few examples, lr=2e-5 overfits. The optimal rate is dataset-size-dependent, and larger datasets tolerate more aggressive rates.

A 40× increase marks a qualitative regime shift.

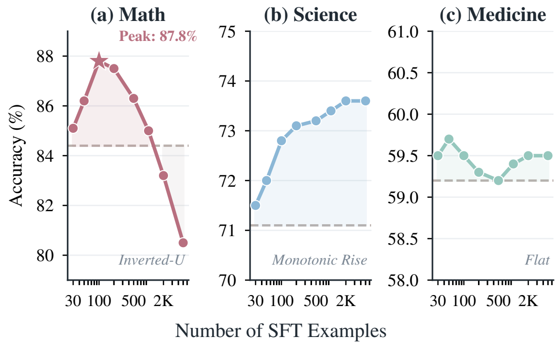

Figure 3: SFT data scaling across three domains (small multiples).

In information-theoretic terms, GRPO with standard lr operates as pure reverse-KL minimization, concentrating probability on existing high-reward modes. Higher learning rates enable distribution shifts that capture new modes, connecting to the framework of Matsutani et al. (2025).

Takeaways

1. The math-optimized GRPO recipe (lr=5e-7) fails across all five domains despite rising training reward.
2. A 40× learning rate increase breaks the mode-seeking trap, yielding +7–25% gains that surpass SFT.
3. On-policy self-distillation provides no improvement, indicating that external signal is necessary.

4.3 SFT Data Scaling: The Inverted-U

SFT effectiveness depends non-trivially on training data amount, and the pattern maps inversely onto base model accuracy (Figure 3). Math shows a sharp inverted-U peaking at N=100 then declining below base. Science rises monotonically. Medicine remains flat regardless of data size. These three patterns correspond to high, moderate, and low base-model accuracy respectively. Full multi-seed results in Appendix F. **Math** exhibits a sharp inverted-U peaking at N=100 then declining below base as excessive demonstrations overwrite correct reasoning. **Medicine** is flat across all data sizes, confirmed by multi-seed validation. Chain-of-thought demonstrations cannot inject medical knowledge the model lacks. **Science** improves monotonically through N=5000 without degradation, its intermediate base leaving room for absorption. The organizing principle is that higher base accuracy means more vulnerability to degradation from excess data. But can this vulnerability be predicted before training begins?

Domain	Base	PPL	Peak Δ	N^*
Math	84.4%	4.34	+3.4%	100
Science	71.1%	2.92	+2.5%	5000+
Medicine	59.2%	2.46	+0.5%	flat

Table 4: SFT data perplexity under the base model correlates with observed scaling behavior. High-perplexity data (Math, 4.34) indicates a large distribution gap, causing early saturation and steep degradation. Low-perplexity data (Medicine, 2.46) is already within the model's distribution, making SFT ineffective regardless of data amount.

4.4 Perplexity Correlates with Degradation

The perplexity of SFT training data under the base model predicts which domains will degrade. We compute perplexity as the mean per-token cross-entropy of the training data, averaged across all tokens using the model's default tokenizer.

The ordering maps directly to degradation severity (Table 4). Math CoT data (ppl=4.34) is most foreign to the model. Science data (ppl=2.92) is moderately distant. Medicine data (ppl=2.46) is closest. High-perplexity data pulls the model further from its natural distribution with each example, causing earlier and steeper collapse. Low-perplexity data aligns with what the model already produces and can be absorbed without destructive interference.

The mechanism follows from forward-KL minimization. SFT expands the model's distribution to cover the training data. When that data is far from the model (high perplexity), expansion overwrites existing capabilities faster. When data is close (low perplexity), expansion is gentle. Perplexity measures the cost of expansion before any training begins. Can we derive a diagnostic with similar explanatory power for GRPO?

4.5 A Pre-Training Diagnostic: Predicting Learning Quality

The learning rate sweep answers whether GRPO will work. A separate question is what *kind* of learning will occur. A single inference-time statistic answers this before training begins. *frac_reward_zero_std* measures the fraction of training prompts where all K=8 sampled responses receive identical reward (all correct or all incorrect), yielding zero advantage variance.

Two regimes emerge. In Medicine (frac_zero_std ≈0.2), 80% of prompts produce mixed results. The model has partial

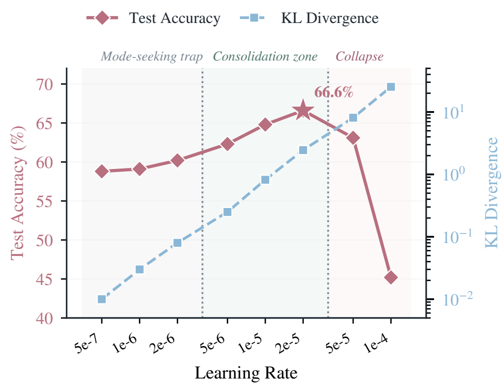

Figure 4: Phase diagram of GRPO learning rate on Medicine, revealing three regimes. **Mode-seeking trap** (lr≤2e-6): accuracy stagnates, KL stays near zero. **Consolidation zone** (5e-6 to 2e-5): accuracy peaks at 66.6%, KL rises moderately. **Collapse** (≥5e-5): KL explodes while accuracy drops sharply. The optimal lr produces a controlled 2.45 nats KL shift, enough for genuine distribution movement without catastrophic drift.

knowledge that manifests unreliably. GRPO stabilizes this unreliable knowledge into consistent correct performance, and the acquired capability transfers to held-out benchmarks (Table 18). The model is not memorizing training answers. It is consolidating latent knowledge that generalizes.

In Math (frac_zero_std ≈0.6), most prompts are already trivially solved. Gradient signal comes from a minority of boundary prompts, and the resulting learning is specific to those prompts. Training reward reaches near-ceiling, yet transfer to MATH-500 is minimal. The optimization learned format, not deeper reasoning.

This makes frac_reward_zero_std a practical decision tool. If low (<0.4), use GRPO with elevated lr for genuine, transferable gains. If high (>0.6), GRPO can improve in-distribution metrics but gains will not generalize. The ordering is monotonic across three domains (Table 18) and theoretically grounded by Bae et al. (2026), who prove that zero-variance groups produce zero gradient regardless of other hyperparameters. Figure 4 visualizes the distinction through KL trajectories. At standard lr, KL stays flat (no distribution shift), while at lr=2e-5, KL increases by an order of magnitude.

4.6 Cross-Domain Transfer

Beyond within-domain OOD transfer, we test whether gains survive a change of domain entirely. If GRPO truly consolidates generalizable reasoning, its advantage should survive domain boundaries.

Transfer	Method	Acc (%)	Std
Math → Science	GRPO	82.6	1.7
	SFT	**84.2**	0.6
	OPD	75.8	–
Med → Science	GRPO	**87.8**	1.2
	SFT	77.9	0.5
Sci → Medicine	GRPO	**77.4**	0.8
	SFT	70.6	0.8
	OPD	67.3	0.2

Table 5: Cross-domain transfer (multi-seed, lr=2e-5). Each row trains on the source domain and evaluates on the target domain's test set. GRPO outperforms SFT on 2/3 domain pairs by +7–10pp. OPD transfers poorly, consistent with the absence of external training signal.

It does. GRPO outperforms SFT on two of three cross-domain pairs (Table 5). GRPO consolidates content-understanding patterns that transfer, while SFT memorizes surface structure of its training demonstrations. The exception (Math→Science) is explained by frac_reward_zero_std, where Math-GRPO learns format rather than transferable reasoning. Direct confirmation of this interpretation comes from Math-trained SFT, which hurts math while improving science by double digits (Appendix N).

5 Discussions

5.1 Forward KL vs Reverse KL: Why Recipes Differ

The results have a clean information-theoretic explanation. SFT minimizes forward KL (mode-covering), so it can inject genuinely new patterns the model has never produced. GRPO approximates reverse KL (mode-seeking), concentrating mass on existing high-reward modes but unable to expand to patterns never sampled. OPD occupies a degenerate position. It trains on the model's own output distribution, providing zero information gain. The model's correct outputs are already within its distribution by construction. Training on them amounts to reinforcing what is already known, while the incorrect outputs that leak through filtering inject noise.

The key consequence is that GRPO can only *consolidate* knowledge the model already partially possesses. It cannot teach new capabilities from scratch. OPD cannot even consolidate, because it lacks the reward pressure that forces the model to distinguish better from worse outputs within its own distribution. Learning rate controls how aggressively GRPO's consolidation occurs. Standard lr barely consolidates. High lr consolidates fully. The *ceiling* of what GRPO can learn is set by what the model can already occasionally produce correctly. The floor, as OPD shows, is the model's existing performance level.

This is why frac_reward_zero_std predicts transfer quality. It measures the fraction of prompts where the model is genuinely uncertain (non-zero advantage variance). Prompts that are always correct or always incorrect contribute zero gradient regardless of learning rate. Medicine (frac=0.2) has 80% uncertain prompts, yielding transferable consolidation. Math (frac=0.6) derives signal from a minority of boundary cases, yielding prompt-specific optimization.

This reconciles our results with (Chen et al., 2026), who show that RLVR cannot expand the solvable frontier. Learning rate controls the *degree* of consolidation within that frontier. Higher rates partially break the mode-seeking constraint by enabling distribution shifts that capture new modes. The SFT→GRPO hybrid exploits this asymmetry. SFT first expands the distribution, then GRPO sharpens within the expanded space. The constraint is that SFT must not over-expand and destroy diversity, which is why N=100 before GRPO outperforms N=5000.

5.2 What Does High-LR GRPO Actually Learn?

The forward/reverse-KL framework makes a testable prediction. Where the model already possesses knowledge, GRPO should learn output *format*. Where knowledge is genuinely absent, aggressive-lr GRPO should acquire *new knowledge* that transfers. The data confirms this (Table 18). On Math (base 84.4%), transfer to MATH-500 is negligible despite near-ceiling training reward. On Medicine, transfer to MMLU-Medical *exceeds* the in-distribution gain. KL divergence confirms the distinction. Medicine shows a large policy shift while Math shows modest growth despite higher final reward.

Two theoretical results ground these observations. Bae et al. (2026) prove that only intermediate-accuracy samples contribute non-zero gradient in group-relative optimization. Our frac_reward_zero_std is the empirical instantiation of this principle. Chen et al. (2026) establish that RLVR improves sampling efficiency without expanding the capability frontier. We extend both to non-reasoning domains. The *degree* of consolidation toward the frontier is lr-dependent. At standard learning rates, GRPO on knowledge domains is not performing reinforcement learning in any meaningful sense (KL $\approx$ 0, no transfer). Only when elevated lr forces genuine policy movement does GRPO consolidate partial knowledge into generalizable performance.

5.3 Practical Recipe Framework

We validate the GRPO lr=2e-5 recipe on four additional architectures from four labs (Table 17). Every model gains substantially on Medicine (+6 to +12pp), regardless of base accuracy, pretraining data, or vocabulary size. The lr sensitivity is consistent across all tested model families in the 7–9B parameter range. A practitioner decision procedure based on base accuracy and data perplexity is given in Appendix A.

6 Conclusion

We show that math-calibrated RLVR does not transfer as a default. The standard GRPO recipe increases training reward but yields little test improvement, exposing a reward–generalization gap under conservative optimization. A 40× learning-rate increase resolves this mismatch, shifting GRPO from prompt-specific sharpening to latent-knowledge consolidation and producing 7–25% gains across multiple-choice domains. These findings reframe RLVR beyond math as a calibration problem: optimization strength must be matched to the domain's reward signal and model uncertainty, rather than inherited from math-oriented defaults.

Limitations

Our analysis is intentionally centered on controlled RLVR calibration: from 7B to 9B instruction-tuned models, LoRA training, and verifiable evaluation. Extending our analysis framework of calibration to larger-scale models, full-parameter updates, and open-ended tasks with less standardized rewards is an important direction for future work.

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A Practical Recipe Guide

Based on our findings, we propose the following decision procedure for practitioners. **High base accuracy (>75%):** GRPO with elevated lr sharpens existing reasoning. SFT helps at very small N but degrades rapidly. **Moderate base (55–75%):** GRPO with lr=5e-6 to 2e-5 breaks the mode-seeking trap. SFT works but hits a lower ceiling. **Low base (<55%):** SFT is generally safer, though GRPO can still produce gains if frac_reward_zero_std indicates learnable signal exists (DeepSeek-7B gains +8pp from base 12.1%). **Data calibration:** Measure SFT data perplexity. High perplexity (>4) indicates use minimal data (N≤100). Low perplexity (<3) allows N≥2000 without degradation.

B Hyperparameters

Parameter	GRPO	SFT
Base model	Qwen2.5-7B-Instruct	
LoRA rank	64	64
LoRA alpha	128	128
Target modules	All linear	All linear
Learning rate	5e-7 to 2e-5	2e-5
Batch size	2 × 4 accum	4 × 4 accum
Training epochs	∼3 (via steps)	3
Num. generations	8	–
Temperature (train)	1.0	–
KL coefficient	0.001	–
Max new tokens	2048 / 1024	–

Table 6: Training hyperparameters for GRPO and SFT. GRPO learning rate is swept across four values (5e-7, 5e-6, 2e-5, 1e-4) per domain. The standard recipe uses 5e-7. Our results show 2e-5 is optimal for knowledge domains.

C SFT Data Scaling Details

Full SFT accuracy curves for Math and Medicine across all data sizes:

Math shows a pronounced inverted-U where accuracy rises +4.2% at N=100 then drops −8.5% below base at N=5000. The 100-to-500 transition (−10.0pp) is particularly sharp, suggesting a phase transition in format overfitting. Medicine has a flatter curve with the peak shifted to N=2000, reflecting its greater need for knowledge injection before saturation.

D Reward Function Details

Math. We extract the final numerical answer and compare against the ground truth after normalization (removing commas, converting fractions to decimals).

N	Math (%)	Medicine (%)
50	87.4	–
100	**88.6**	59.5
200	86.9	–
500	78.6	59.5
1000	–	58.6
1500	–	60.2
2000	–	**61.7**
5000	75.9	57.8

Table 7: SFT accuracy at different data sizes. Math peaks at N=100 with steep degradation beyond. Medicine peaks at N=2000 with gentler slopes. Base accuracy: Math 84.4%, Medicine 59.2%.

MCQ domains (Science, Medicine, Law, Commonsense). We extract the letter answer (A/B/C/D) from the model's response and compare against the gold label. Reward is 1 for exact match, 0 otherwise.

E SFT Data Construction

For domains without naturally occurring chain-of-thought solutions (Science, Medicine, Law, Commonsense), we generate demonstrations via rejection sampling from the base model. For each question, we sample 8 responses at temperature 0.7 and keep the shortest response that produces the correct final answer. Questions where all 8 attempts fail are excluded from the SFT training set. This ensures data quality without relying on a stronger teacher model, maintaining fairness in our comparison.

For Math, we use the step-by-step solutions provided in GSM8K.

F Full SFT Data Scaling Results

Multi-seed SFT accuracy across all domains is in Tables 8–10. Math shows a sharp inverted-U robust across 4 seeds. Medicine is flat regardless of data size, so GRPO is the only effective method. Science improves monotonically.

G GRPO Learning Rate Sweep Details

The complete GRPO learning rate sweep across all domains is in Table 11. The optimal learning rate depends on dataset size. MCQ domains with 2000 training examples benefit most from lr=2e-5, while Law (31 examples) peaks at lr=5e-6.

For GRPO at standard lr=5e-7, we ran multi-seed validation on Math (s42: 84.7%, s123:

N	s42	s123	s456	Avg
50	87.4	–	–	87.4
100	87.8	88.6	87.6	**87.8**
200	86.9	86.3	85.4	86.2
500	78.6	79.1	79.8	79.2
2k	77.6	79.2	79.2	78.7
5k	75.9	77.6	77.8	77.1

Table 8: Math SFT (%). Base: 84.4. Peak at N=100, steep degradation beyond.

N	s42	s123	s456	Avg
100	59.5	60.0	60.6	59.7
500	59.5	58.3	60.9	59.3
1k	58.6	59.5	–	59.1
2k	61.7	58.5	59.5	59.4
3k	59.2	59.3	57.9	58.8
5k	58.7	–	58.5	58.6

Table 9: Medicine SFT (%). Base: 59.2. Flat at all N. s42 peak is noise (σ=1.5).

N	s42	s123	s456	Avg
100	71.7	72.5	71.6	71.9
500	71.8	73.6	71.9	72.4
1k	72.5	–	72.8	72.6
2k	73.4	73.1	72.7	73.1
5k	73.8	73.6	74.0	**73.6**

Table 10: Science SFT (%). Base: 71.1. Monotonic improvement, no degradation.

Three domain-specific SFT scaling patterns emerge from multi-seed validation. Math shows the sharpest inverted-U (peak at N=100, $-$10.7pp by N=5k), driven by format memorization overwriting correct reasoning. Medicine remains flat ($\pm$0.6pp) regardless of data volume: chain-of-thought demonstrations cannot inject medical knowledge the base model lacks. Science improves monotonically (+2.5pp from N=100 to N=5k), suggesting the model benefits from structured science demonstrations without overfitting risk.

Learning Rate	**Math**	**Science**	**Medicine**	**Law**	**Pattern**
Base	84.4	71.1	59.2	57.6	–
5e-7 (standard)	84.7	71.3	58.8	54.5	Flat/hurt
5e-6	–	73.9	61.3	**58.9**	Moderate
1e-5	–	–	–	58.8	–
2e-5	**92.2**	**79.1**	**66.6**	56.4	Strong/overfit
1e-4	–	–	0.0	–	Collapsed
Δ (best vs base)	+7.8	+8.0	+7.4	+1.3	
Δ (best vs SFT)	+4.4	+5.5	+7.1	+0.1	

Table 11: GRPO accuracy (%) across learning rates and domains. MCQ domains (Math, Science, Medicine) peak at lr=2e-5. Law (binary reward, 31 examples) peaks at lr=5e-6. The optimal lr is dataset-size dependent.

85.4%, average: 85.0%) and Science (s42: 71.3%, s123: 71.7%, average: 71.5%). The near-zero improvement is consistent and not a single-seed artifact. The average test delta at standard lr is +0.6% (Math) and +0.4% (Science), both within noise.

H Cross-Model Validation: Qwen3-8B

To validate that our findings are not specific to Qwen2.5-7B-Instruct, we replicate key experiments on Qwen3-8B, a newer-generation model with different pretraining data and architecture. The complete results are in Table 12.

GRPO lr=2e-5 outperforms SFT on both model generations across all three domains (Table 12). The pattern holds regardless of architecture generation or pretraining corpus. Qwen3-8B shows proportionally larger Medicine gains (+12.4%) due to its lower base accuracy (50.5%) providing more headroom for knowledge consolidation.

I Training Dynamics: Reward Trajectories

Key reward statistics during GRPO training on Medicine at different learning rates are in Table 13.

	Math	**Science**	**Medicine**
Qwen3-8B			
Base	81.5	73.9	50.5
SFT best	82.9_{5k}	73.4_{5k}	62.1_{500}
GRPO lr=2e-5	**90.5**	**78.7**	**62.9**
Δ best vs base	+9.0	+4.8	+12.4
Qwen2.5-7B-Instruct			
Base	84.4	71.1	59.2
SFT best	87.8_{100}	73.6_{5k}	59.5_{100}
GRPO lr=2e-5	**92.2**	**79.1**	**66.6**
Δ best vs base	+7.8	+8.0	+7.4

Table 12: Cross-model validation across three domains. GRPO lr=2e-5 outperforms SFT on both model generations for all three domains. On Qwen3-8B, Math GRPO reaches 90.5% (+9.0%), Science 78.7% (+4.8%), and Medicine 62.9% (+12.4%), all surpassing SFT.

These numbers quantify the training dynamics visualized in Figure 2.

Metric	lr=5e-7	lr=5e-6	lr=2e-5	lr=1e-4
Initial reward	0.50	0.50	0.50	0.50
Final reward	0.53	0.55	0.83	0.12
Peak reward	0.53	0.55	0.83	0.68
Δ reward	+0.03	+0.05	+0.33	−0.38
Test Δ	−0.4%	+2.1%	+7.4%	collapsed
frac_zero_std (final)	0.40	0.45	0.62	–

Table 13: GRPO training dynamics on Medicine. lr=2e-5 raises reward to 0.83 (+7.4% test). lr=5e-7 is flat.

Category	Models	GPU-hrs	%
SFT (all domains)	55	22	17%
GRPO standard lr	15	18	14%
GRPO lr sweep	10	12	9%
GRPO lr=2e-5 (seeds)	10	14	11%
SFT→GRPO hybrids	4	5	4%
Cross-model (Qwen3)	10	12	9%
Cross-model (Mistral/Yi/DS)	8	18	14%
KL/K ablation	6	8	6%
Evaluation (vLLM)	–	18	14%
Failed/exploratory	82	29	22%
Total	**200+**	**∼200**	100%

Table 14: Compute budget. ∼200 H200 GPU-hours across 200+ models. LoRA kept memory below 35GB, enabling single-GPU parallelism.

J SFT→GRPO Hybrid Details

The full SFT→GRPO hybrid results on Math are in Table 15, varying the amount of preceding SFT data.

Pipeline	s42	s123	Avg	Δ
Base	84.4	84.4	84.4	–
SFT(n=100) only	87.8	88.6	88.2	+3.8
GRPO lr=2e-5 only	92.8	–	92.8	+8.4
SFT(n=100)→GRPO	89.8	88.7	89.3	+4.9
SFT(n=5000)→GRPO	82.9	–	82.9	−1.5

Table 15: SFT→GRPO hybrid results on Math (%, seed 42). Pure GRPO at lr=2e-5 achieves 92.8% (seed 42, multi-seed avg 92.2%), outperforming both the hybrid SFT(100)→GRPO (89.3%, 2-seed avg) and SFT alone (88.2%). Excessive SFT (n=5000) before GRPO degrades below base because over-SFT destroys the diversity GRPO needs.

The interaction between SFT amount and subsequent GRPO effectiveness is simple. Preceding SFT data controls GRPO's exploration capacity. With n=100, the model learns the response format without losing output diversity, providing GRPO with the variance it needs for contrastive learning. With n=5000, the model has collapsed to a narrow distribution that mimics training demonstrations, leaving GRPO with homogeneous response groups and near-zero advantage variance.

K KL Penalty and Group Size Ablation

To verify that learning rate is the primary variable (rather than an interaction artifact with KL penalty or group size), we ablate β and K while holding lr=2e-5 fixed on Medicine.

The pattern is monotonic: stronger KL penalty → less distribution shift → smaller test gain. KL regularization constrains how far the policy can move from the reference. At standard lr, the movement is already near-zero regardless of β. High lr breaks this constraint, but increasing β partially reinstates it. Removing KL entirely at standard lr (β=0) makes performance *worse* (57.0%, below base). KL stabilizes training when learning rate is too low to provide directional gradient signal. Practitioners should use low β (0.001 or less) with elevated learning rates.

L Cross-Model Validation

An additional pattern emerges. Gain magnitude does not simply increase with lower base accuracy. Yi-1.5 (base 43.3%) gains less (+6.0%) than Mistral (base 34.3%, +8.5%). The effective learning signal depends on the model's internal uncertainty distribution (frac_reward_zero_std), not raw accuracy alone.

M Out-of-Distribution Transfer

The transfer ratio column crystallizes the core insight. When GRPO training signal comes primarily from genuine model uncertainty (Medicine, low frac_zero_std), the resulting learning is domain knowledge that transfers to any test of the same knowledge. When signal comes from a minority of boundary cases (Math, Commonsense, high frac_zero_std), optimization is specific to those cases and does not generalize. This distinction between knowledge acquisition and format optimization is not visible from ID evaluation alone. It requires an OOD probe to reveal.

N SFT Learns Format, Not Domain Knowledge

A direct test of the format-vs-knowledge hypothesis. We train SFT on Math-only data (MATH-500, Qwen2.5-7B-Instruct, 3 seeds) and evaluate on unrelated domains.

Setting	Accuracy (%)	Δ vs base	Interpretation
lr=5e-7, β=0.001 (std)	58.8	−0.4	Standard recipe fails
lr=5e-7, β=0 (no KL)	57.0	−2.2	Removing KL → worse (drift)
lr=2e-5, β=0.001	**66.6**	**+7.4**	Optimal (paper default)
lr=2e-5, β=0.01	62.4	+3.2	Stronger KL reduces gain
lr=2e-5, β=0.1	60.2	+1.0	Heavy KL nearly eliminates gain
lr=2e-5, K=4	62.9	+3.7	Fewer samples → noisier advantage

Table 16: KL penalty (β) and group size (K) ablation on Medicine. Increasing β from 0.001 to 0.1 progressively constrains policy shift, reducing gain from +7.4% to +1.0%. Removing KL entirely at standard lr (β=0, first row) causes *degradation* below base: KL stabilizes training when lr is too low to provide directional signal. Learning rate remains the primary lever in all settings.

Model	Lab	Base (%)	GRPO lr=2e-5 (%)	Δ	Observation
Medicine (MedQA, 1273 test)					
Qwen2.5-7B-Instruct	Alibaba	59.2	**66.6**	+7.4	Primary model
Qwen3-8B	Alibaba	50.5	**62.9**	+12.4	Newer generation
Yi-1.5-9B-Chat	01.AI	43.3	**49.3**	+6.0	Llama architecture
Mistral-7B-Instruct	Mistral AI	34.3	**42.8**	+8.5	European model
DeepSeek-7B-chat	DeepSeek	12.1	**20.3**	+8.2	2023 model (weak base)
Science (ARC-Challenge, 5413 test)					
Qwen2.5-7B-Instruct	Alibaba	71.1	**79.1**	+8.0	Primary model
Qwen3-8B	Alibaba	73.9	**78.7**	+4.8	Higher base → less headroom
Mistral-7B-Instruct	Mistral AI	61.2	**70.4**	+9.2	Lower base → larger gain

Table 17: Cross-model validation of GRPO lr=2e-5 across five architecturally distinct model families from four research labs. All models show positive improvement (+6 to +12pp on Medicine, consistent multi-point gains on Science). The absolute gain is consistent across models regardless of base accuracy, showing that the lr sensitivity phenomenon holds across all tested architectures. Models span 2023–2025 releases with different pretraining corpora, vocabulary sizes, and attention mechanisms.

Math-trained SFT *degrades* math performance (−17.8pp) while dramatically improving science (+12.6pp) and general knowledge (+7.1pp). The model does not acquire mathematical reasoning from SFT. Instead, it learns the structured MCQ answer format that transfers to all domains. This confirms our forward-KL interpretation: SFT expands general output coverage rather than injecting domain-specific capability, explaining why SFT shows cross-domain transfer advantages in Table 5.

O On-Policy Self-Distillation (OPD) Results

OPD trains the model on its own filtered correct outputs (selecting correct responses from K=8 samples at temperature 0.7), without any external teacher or reward signal. Results are in Table 20.

Domain	OOD Benchmark	Base	GRPO	Δ OOD	Δ ID	Transfer Ratio	frac_zero_std
Medicine	MMLU-Medical	47.2	55.8	+8.6	+7.4	1.16×	0.2
Math	MATH-500	72.8	74.2	+1.4	+7.8	0.18×	0.6
Commonsense	WinoGrande	74.1	74.8	+0.7	+25.0	0.03×	0.7

Table 18: Out-of-distribution transfer evaluation. Transfer Ratio = ΔOOD / ΔID measures how much of the in-distribution gain generalizes. Medicine achieves near-complete transfer (ratio >1), meaning genuine knowledge consolidation. Math and Commonsense show minimal transfer despite large ID gains, meaning format-specific optimization. The frac_reward_zero_std diagnostic at training onset predicts this ordering: lower values correspond to stronger OOD transfer, consistent with the theoretical prediction of Bae et al. (2026).

Benchmark	Base (%)	SFT-Math (%)	Δ
Math-500	60.2	42.4±0.2	−17.8
ARC-Challenge	74.9	87.5±0.1	+12.6
MMLU	61.1	68.2±0.1	+7.1

Table 19: SFT trained on math data hurts math but improves unrelated domains. The model learns MCQ output format (selecting A/B/C/D cleanly), not mathematical reasoning. 3 seeds, std < 0.2 on all benchmarks.

	Science	Medicine
Base	71.1	59.2
OPD (5 seeds)	71.7±0.2	60.0±0.9
Δ	+0.6	+0.9
Filtering rate	69.1%	53.7%

Table 20: On-policy self-distillation (OPD) results on Science and Medicine. OPD produces no meaningful improvement over base ($\Delta < 1$pp in both domains). The null result indicates that self-generated data, without external signal from a teacher (SFT) or reward function (GRPO), cannot push the model beyond its current capability boundary.